

Exam-2

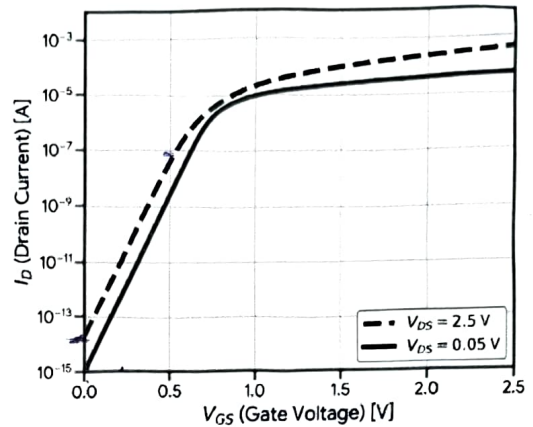
EE226 Semiconductor Devices

Total Marks: 40

Q1. From the figure given below:

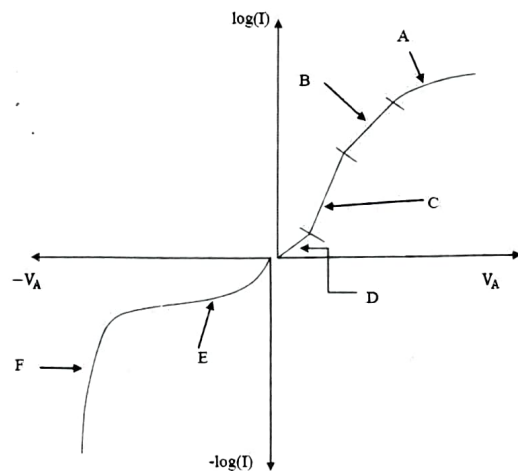
(6 Marks)

- Annotate the subthreshold region, linear and saturation region (2)
- Which region can be used for switch operation i.e., ON and OFF region (1)
- Which region can be used for amplifier operation (1)
- What changes are observed by increasing drain voltage to 2.5V? Explain the observations. (2)
- Calculate the subthreshold slope. (2)



Q2. For the non-ideal diode I-V shown in figure, name the different regions of operation. Reason each operation briefly.

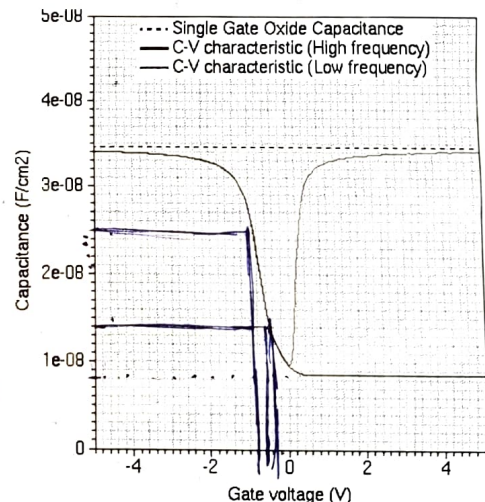
(6 Marks)



Q3. For a MOSCAP with SiO_2 as oxide and Si as semiconductor, a CV measurement is shown. Answer the following:

(8 Marks)

- Calculate the Oxide thickness. (1)
- What is the type of the semiconductor? (1)
- Calculate the doping of the semiconductor. (2)
- Calculate the Threshold voltage V_{th} (Assume work function of metal and semiconductor are equal i.e. $\phi_m = \phi_{Si}$) (2)
(Calculate V_{th} analytically. Just writing value directly from the plot will not carry any marks.)
- Calculate flatband capacitance, flatband voltage, and the actual threshold voltage (2)



Q4. Answer the following questions:

(10 Marks)

Note: You need to show the complete I-V curve until the device burns/damages. No limit (Current or Voltage) is applied to the power supply.

- Plot I-V for a resistor. Keep increasing the voltage until the device burns out. Draw and explain the effect in different part of the I-V. (2)
- Plot I-V for a diode in a linear scale. Keep increasing the voltage until the device burns out. Draw and explain the effect in different part of the I-V. (2)
- Plot I_{ds} vs V_{gs} plot in a linear scale with small V_{ds} . Keep increasing the voltage until the device burns out. Draw and explain the effect in different part of the I-V. (3)
- Plot I_{ds} vs V_{ds} plot in linear scale with $V_{gs} > V_{th}$. Keep increasing the voltage until the device burns out. Draw and explain the effect in different part of the I-V. (3)

Relevant parameters and formulas

$$\epsilon_{Si} = 12, \epsilon_{SiO_2} = 4, q = 1.6 \times 10^{-19} C, \epsilon_0 = 8.845 \times 10^{-14} F/cm, k = 8.6 \times 10^{-5} \frac{eV}{K},$$

$$V_T = \frac{kT}{q} = 0.026 V @ T = 300K, \mu_n = 1500 \frac{cm^2}{Vs}, n_i \text{ for Si} = 1.1 \times 10^{10} / cm^3$$

$$n_i = \sqrt{N_c N_v} e^{-\frac{E_g}{2kT}}$$

$$I = I_0 \left(e^{\frac{V_A}{\eta V_T}} - 1 \right) \quad \eta \text{ is ideality factor}$$

$$E_i - E_F = \phi_B = \frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right), \text{ Depletion width } W = \sqrt{\frac{2\epsilon_{Si}\psi_s}{qN_A}},$$

ψ_s is the surface potential i.e. voltage drop in silicon ($=V_{si}$)

$$C_{ox} = \frac{\epsilon_{ox}\epsilon_0 A}{t_{ox}}, -Q_{dep} = C_{ox}V_{ox}, Q_{dep} = -qN_A W$$

$$\text{Debye length} = L_D = \sqrt{\frac{\epsilon_{Si}kT}{q^2 N_{doping}}}$$